

Typed Falsification-Aware Scientific Authority as a Least Fixed Point

Paper D — hardened manuscript V2 Scientific cut: D1 parent, R2 powerset-license calculus, and three scientific cases

Workflow cut: `academic-paper-skills@188e83e63957` (full digest in `manuscript/BUILD_PROVENANCE.json`)

Abstract

A falsifier should retract claims by derivation and by evidence type. Boolean dependency graphs solve only the first problem: they can say whether a claim remains derivable, but not whether the surviving derivation licenses a theorem, a finite exact statement, a prospective claim, or a post-outcome repair. We introduce a finite typed authority calculus for positive conjunctive scientific rule systems. Let $\mathbf{Lambda}$ be a finite universe of authority licenses and label every claim by a subset of $\mathbf{Lambda}$. Independent seeds carry declared licenses. Each rule has an explicit cap and may transmit only the intersection of its premises' licenses with that cap. Directly refuted claims are fixed at the empty label. The monotone operator

$F_R(x)_q = \text{empty for } q \text{ in } R,$

and otherwise the union of the seed label and all capped rule transfers into q , has a least fixed point on the finite powerset lattice.

We prove finite convergence, rule-order independence, and a typed proof-tree theorem: a license reaches a claim exactly when some finite untainted proof tree carries that license in every leaf seed and every rule cap. Unsupported cycles remain bottom; seeded cycles may propagate; adding refutations can only remove licenses; alternative derivations preserve exactly their surviving licenses; and the removed claim-license pairs form the unique minimal typed retraction within the declared license algebra. The cap mechanism internalizes nonpromotion. A post-outcome repair rule cannot manufacture prospective authority, and bounded computational evidence cannot manufacture theorem authority even when its untyped claim is reachable.

Three scientific cases demonstrate the specialization: forecast falsification, query-specific information falsification, and bounded-computation nonpromotion. A reusable domain-agnostic evaluator independently re-derives the committed verdicts, and an external X.509 trust-store instantiation shows that the target merge obstruction is non-vacuous: 46 hybrid cases occur among 1,962 third-party OpenSSL-derived merge tasks. The evaluator's zero unsafe merges and zero needless rejections are not detector-performance measurements; because it authorizes exactly the parent-authorized set, those values are analytic consequences of the formal semantics. The empirical contribution is obstruction occurrence and the method-dependent cost paid by naive merge

policies. Truth-maintenance systems, positive Datalog, annotated/semiring provenance, minimal supports, causality, and deletion robustness are direct donors. The residual contribution is the scientific evidence-license specialization, cap-preserving nonpromotion, reusable evaluator, and externally instantiated authority/retraction behavior.

1. Introduction

Scientific records mix claims supported in different ways:

- analytic theorem;
- constructive bound;
- finite exact computation;
- prospective prediction;
- forecast only;
- post-outcome repair;
- bounded computation pending external replay.

A single counterexample may invalidate one layer but not another. An untyped dependency graph can preserve independent derivations, yet it can still overpromote a surviving repair. If a post-outcome predictor becomes exact on the observed panel, it should not inherit prospective authority merely because the old prospective claim and the repair share a conclusion string.

This paper adds evidence licenses to the least-fixed-point calculus. The mathematical substrate is deliberately close to positive Datalog and provenance: finite claims, positive conjunctive rules, monotone iteration, and finite proof trees. The new requirement is scientific nonpromotion—derivability and license must travel together.

1.1 Contributions

1. **Powerset-license semantics:** claim authority is a subset of a finite license universe.
2. **Capped transfer:** a rule transmits only licenses present in all premises and permitted by the rule.
3. **Typed proof-tree characterization:** fixed-point licenses are exactly the licenses of finite untainted proof trees.
4. **Typed retraction within the declared algebra:** refutation is point-wise monotone and removes exactly the unsupported claim-license pairs.
5. **Internal nonpromotion:** prospective and theorem licenses cannot appear without a fully licensed proof tree.
6. **Three scientific cases:** forecast falsification, query-specific information falsification, and bounded-computation nonpromotion.
7. **Reusable and external instantiation:** a domain-agnostic evaluator rederives the committed cases, and third-party X.509 material establishes that the typed-merge obstruction occurs outside the originating paper corpus.

Items 1–4 deliberately reuse donor mathematics. The paper does not claim to invent fixed-point evaluation, provenance, minimal-support reasoning, or deletion robustness; its residual is the scientific evidence typing and nonpromotion specialization, plus the executable and external instantiations.

2. Claims, licenses, and rules

Let Q be a finite claim set and Λ a finite license universe. Example licenses include

THEOREM, CONSTRUCTIVE_BOUND, FINITE_EXACT, PROSPECTIVE, FORECAST_ONLY, POST_OUTCOME, BOUNDED_COMPUTATION, EXTERNAL_REPLAY.

A label is a subset of Λ , ordered by inclusion. Join is union, bottom is the empty set.

Each claim q has an independent seed label $\sigma(q) \subseteq \Lambda$. Each positive rule is a triple

$r = (A \rightarrow h, K_r)$,

where $A \subseteq Q$ is finite, $h \in Q$, and $K_r \subseteq \Lambda$ is the rule cap.

For premise labels ℓ_a , define

$\tau_r((\ell_a)_{a \in A}) = K_r \cap \bigcap_{a \in A} \ell_a$.

A license can pass a rule only if every premise carries it and the rule cap permits it. A zero-antecedent rule is equivalent to an independent seed and is omitted from the core presentation.

Let $R \subseteq Q$ be directly refuted claims.

3. Typed least-fixed-point semantics

For a label assignment $x \in (2^\Lambda)^Q$, define

$F_R(x)_q = \text{empty}$ if $q \in R$,

and otherwise

$F_R(x)_q = \sigma(q) \cup \bigcup_{r: \text{head}(r)=q} \tau_r(x|_{\text{body}(r)})$.

F_R is monotone. Starting from all-empty labels, iterate until stabilization and define

$\text{Auth}_\Lambda(R) = \text{lfp}(F_R)$.

Theorem 1 (finite convergence and order independence). The iteration converges after at most $|Q||\Lambda|+1$ strict license additions. Every fair rule order reaches the same least fixed point.

Proof. Labels only gain licenses during an iteration with fixed R . There are at most $|Q| \cdot |\Lambda|$ claim-license pairs. The operator, not the scan order, defines the least fixed point. ■

4. Typed proof trees

A finite proof tree for (q, λ) is valid under R when:

1. its root claim is q and no node claim is directly refuted;
2. a leaf a has $\lambda \in \text{sigma}(a)$;
3. an internal node applies a registered rule $A \rightarrow h$ whose cap contains λ , and each antecedent has a child proof tree carrying λ .

Theorem 2 (typed proof-tree equivalence). License λ belongs to $\text{Auth_Lambda}(R)_q$ if and only if there exists a finite untainted proof tree for (q, λ) .

Proof. Induct on the first iteration round in which the license appears to construct a tree. Conversely, induct on tree height to show that every leaf license enters from a seed and every internal license passes its capped rule. ■

An untyped authorization set is recovered by retaining claims with nonempty labels, but the typed object contains strictly more information.

5. Cycles and nonpromotion

5.1 Unsupported cycles

Rules $a \rightarrow b$ and $b \rightarrow a$ with no seed labels have all-empty least fixed point. A cycle is a derivation shape, not evidence.

If a has a theorem seed and both rule caps permit **THEOREM**, the license propagates to b . If one cap permits only **FINITE_EXACT**, no theorem license crosses that edge.

5.2 License conservation

Corollary 3. Every license at a conclusion appears in every leaf seed and every rule cap on at least one finite proof tree.

Thus the semantics cannot invent a stronger evidence class.

5.3 Post-outcome repair

A repair rule capped by

`{THEOREM, FINITE_EXACT, POST_OUTCOME}`

cannot transmit **PROSPECTIVE**, even if a premise happened to carry both. This makes “repair is not prospective confirmation” a theorem of the registered authority model rather than a prose convention.

6. Typed retraction

Theorem 4 (refutation monotonicity). If $R \text{ subseteq } R'$, then for every claim

$\text{Auth_Lambda}(R')_q \text{ subseteq } \text{Auth_Lambda}(R)_q$.

Proof. $F_{R'}(x)$ is pointwise contained in $F_R(x)$ for every x ; monotone iteration from bottom preserves containment. ■

Alternative derivations preserve the union of licenses carried by their surviving proof trees.

Define the pre-falsifier label assignment $A_{\text{pre}} = \text{Auth_Lambda}(\text{empty})$ and post-falsifier assignment $A_{\text{post}} = \text{Auth_Lambda}(R)$. The typed retraction is

$\text{Ret_Lambda}(R) = \{(q, \text{lambda}) : \text{lambda} \text{ in } A_{\text{pre}}(q) \setminus A_{\text{post}}(q)\}$.

Theorem 5 (minimal typed retraction in the declared algebra). A_{post} retains every and only claim-license pair with a finite untainted proof tree under the registered seeds, rules, caps, and refutations. Hence $\text{Ret_Lambda}(R)$ removes exactly the unsupported pairs and no supported pair.

This is semantic minimality relative to the declared authority algebra. The generic ideas of dependency-directed retraction, annotated provenance, minimal supports, and deletion robustness are donor-owned; the theorem is used here to make the scientific license policy executable rather than claimed as a new general retraction theory.

7. Case I: TARE forecast falsification

The original TARE forecaster combined:

- explicit feasible constructions, licensing a constructive upper bound;
- an independent all-size support-two theorem;
- a compact equality and regime label supported by finite/prospective rows.

A fresh exact row gives $C_{\text{DP}} = 10 < F = 11$. Direct refutation removes universal closed-form exactness and the old regime label. It does not remove the feasible upper-bound license or the independent support theorem.

A repaired support-two forecaster may receive **THEOREM** from the independent support theorem and **POST_OUTCOME** from its construction, but its repair rule cap excludes **PROSPECTIVE**. The original denominator remains 9,545/9,546; the repaired panel is a new evidence object.

8. Case II: decision survives value and witness falsifiers

Paper C proves a four-index unary-optimality decision theorem. Separate exact constructions show that complete pair information fails to determine exact

value or triple-block presence, and that all proper interaction marginals fail to determine exact value.

The decision claim and value/witness claims therefore occupy separate nodes with separate rules. Refuting pair-value sufficiency removes its licenses but leaves the decision theorem license untouched. Both members of each counterexample pair make the same decision.

This case demonstrates query typing: the same representation can have **THEOREM** authority for a decision query and no exact-value authority.

9. Case III: bounded computation cannot decide D_4

The non-quantum programme has analytic licenses for the one-unit generalized-Davenport corridor and saturation-defect lemma. An exact internal support-frontier computation reports no length-31 obstruction through support 22, but its own metadata explicitly denies theorem authority and requires external replay.

Assign that frontier licenses

`{BOUNDED_COMPUTATION, EXTERNAL_REPLAY}`.

Any rule from the frontier to “support at least 23” is capped by the same set. No proof tree carries **THEOREM** to exact $D_4(C_5^3)$ or 31 in $C_0(C_5^3)$. Those claims remain unlicensed at theorem level.

The case is a practical control against a common publication error: repeated internal exact implementations do not automatically become independent mathematical replication.

10. Reusable evaluator and external X.509 instantiation

The calculus is implemented in `packages/typed-merge-evaluator/` as a domain-agnostic schema and deterministic evaluator. The package does not import the paper-specific **ORION** implementation. It represents authority systems as seeds, capped rules, refutations, and expected verdicts; computes the least fixed point and untainted proof-tree authorization; and includes regression instances for the committed Cedar and X.509 domains. Thirty-two package tests re-derive the committed verdicts and re-verify the bound source digests.

The external instantiation uses third-party OpenSSL 3.6.4 test certificates. Across 1,962 real trust-store merge tasks, 46 tasks satisfy the registered hybrid condition—authorization under the flat union but not under either parent-authorized result. The empirical conclusion is therefore limited but useful: the obstruction targeted by the calculus occurs in deployed-style certificate material rather than only in synthetic scientific examples.

The round-2 metrics require a strict analytic/empirical separation. Let

`parent := v_A or v_B`

`and`

`hybrid := v_union and not parent.`

The registered `M5_TYPED_WITNESS` decision is exactly `parent`. Consequently,

`unsafe_merges[M5] = parent and hybrid = false`

`and`

`needless_rejections[M5] = (not parent) and parent = false`

by propositional identity. The reported M5 precision and recall of 1.0 therefore do **not** measure detector performance on X.509; they follow from D2-C2 and the benchmark definitions for any corpus.

The quantities with genuine empirical content are instead the obstruction prevalence and the behavior of alternative policies. In the `PARITY_PARTITION` family, flat union performs 4 unsafe merges; intersection and reject-all incur 63 needless rejections each; the `M4_OURS_B` baseline incurs 14. The corpus also satisfies the registered non-trivial checks `c3_violations = 0`, `c4_resurrections = 0`, and `c4_upstream_mirrors_ok = true`. These observations could have differed across material and therefore remain measurements.

This application is not a security claim: no attack or threat model is evaluated. It is an external-domain existence-and-cost instantiation of the typed authority semantics.

11. Relation to prior work

Doyle's truth-maintenance system, assumption-based truth maintenance, and belief-revision models own dependency-directed update. Positive Datalog owns least-fixed-point rule semantics. Semiring and annotated provenance own typed query annotations, alternative derivations, recursion, and deletion behavior. Recent recursive-Datalog causality work organizes minimal supports as hypergraphs and derives causes, responsibility, and deletion robustness; the 2026 stratified extension shows that negation changes this structure substantially.

The paper therefore claims no generic novelty for fixed points, proof trees, semiring labels, minimal supports, hitting sets, causality, or deletion robustness. A current hostile novelty subtraction also treats the generic minimality content of Theorem 5 as donor-adjacent rather than as a standalone novelty claim. The residual is deliberately smaller:

- a finite scientific evidence-license vocabulary;
- cap-preserving rule semantics that encodes nonpromotion between evidence classes;
- typed retraction instantiated on falsified/repared scientific records;
- a reusable evaluator for that specialization;

- an external X.509 instantiation showing non-vacuous obstruction occurrence and method-dependent merge costs.

12. Reproducibility

The R2 verifier represents licenses as bitsets, checks unsupported and seeded cycles, prospective and theorem caps, and refutation monotonicity. For hundreds of finite random systems, iterative evaluation is compared against exhaustive enumeration of all fixed points to identify the least one. The proofs carry authority only at the scope stated in the claim ledger.

The reusable evaluator serializes claims, seed licenses, rules, caps, and refutations in a public schema and exposes a deterministic evaluator. Its examples re-encode the committed Cedar and X.509 cases, and its tests re-verify the round-1 receipt digests and all 268 round-2 source bindings before re-deriving the verdicts.

The three scientific cases bind existing committed evidence objects. The X.509 application is separately bound to the third-party certificate material and must retain the analytic/empirical distinction described in Section 10.

13. Limitations

1. Rules are positive and conjunctive; stratified negation, defaults, probabilistic evidence, and inconsistency are out of scope.
2. The license universe and caps are curated scientific policy, not inferred automatically.
3. Powerset intersection is one useful distributive transfer algebra, not a universal authority algebra.
4. The case studies are scientific/compiler records, not a human-subject usability evaluation.
5. Generic fixed-point, provenance, minimal-support, and deletion-robustness mathematics is donor-owned.
6. The X.509 study establishes obstruction occurrence and baseline costs, not detector accuracy, security, or cross-institution deployment.
7. M5's zero unsafe merges, zero needless rejections, and perfect precision/recall are analytic identities under the registered definitions and must not be interpreted as measured performance.

14. Discussion and conclusion

Derivability alone is not scientific authority. A claim can remain derivable under a weaker license, or a repaired claim can regain exactness without regaining prospective status. The typed least fixed point makes those distinctions machine-checkable.

The calculus is fail-closed in two dimensions. Unsupported cycles cannot create claims, and underlicensed derivations cannot create stronger evidence types. At

the same time it is minimally destructive within the declared authority algebra: independent proof trees preserve exactly the licenses they still carry.

The external application narrows rather than inflates the empirical claim. It shows that the obstruction is not synthetic-only and that naive merge policies pay different measurable costs. It does not convert a definitional invariant into detector evidence, and it does not turn donor-owned provenance machinery into a novelty claim. That separation is the intended use of the calculus itself.

Selected references

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- C. Bourgaux, P. Bourhis, L. Peterfreund and M. Thomazo, *Revisiting Semiring Provenance for Datalog*, KR 2022, DOI 10.24963/kr.2022/10.
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- R. B. Thapa and S. Staab, *Causality and Minimal Supports in Recursive Datalog*, arXiv:2607.16443 (2026).
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Publication decision record

Primary target: Journal of Automated Reasoning.

Fallback: ACM Transactions on Computational Logic.

Current posture: bounded formal-methods paper with reusable implementation and external-domain instantiation; no broad-AI, security, or detector-superiority claim.

Completed: formal claim ledger; reusable evaluator and regression suite; external X.509 domain; independent in-repo reproduction of committed verdicts; round-2 analytic/empirical reframe; current hostile novelty subtraction.

Open filing/package items: target-format manuscript/package, figures if required by venue, exact submission manifest, cover letter, archive/licence, and human filing metadata. External peer review remains external by definition.

Stop rule: no further theory recursion merely to manufacture novelty. Any future scientific expansion must introduce a genuinely new externally testable question rather than relabel donor-owned provenance/retraction machinery.